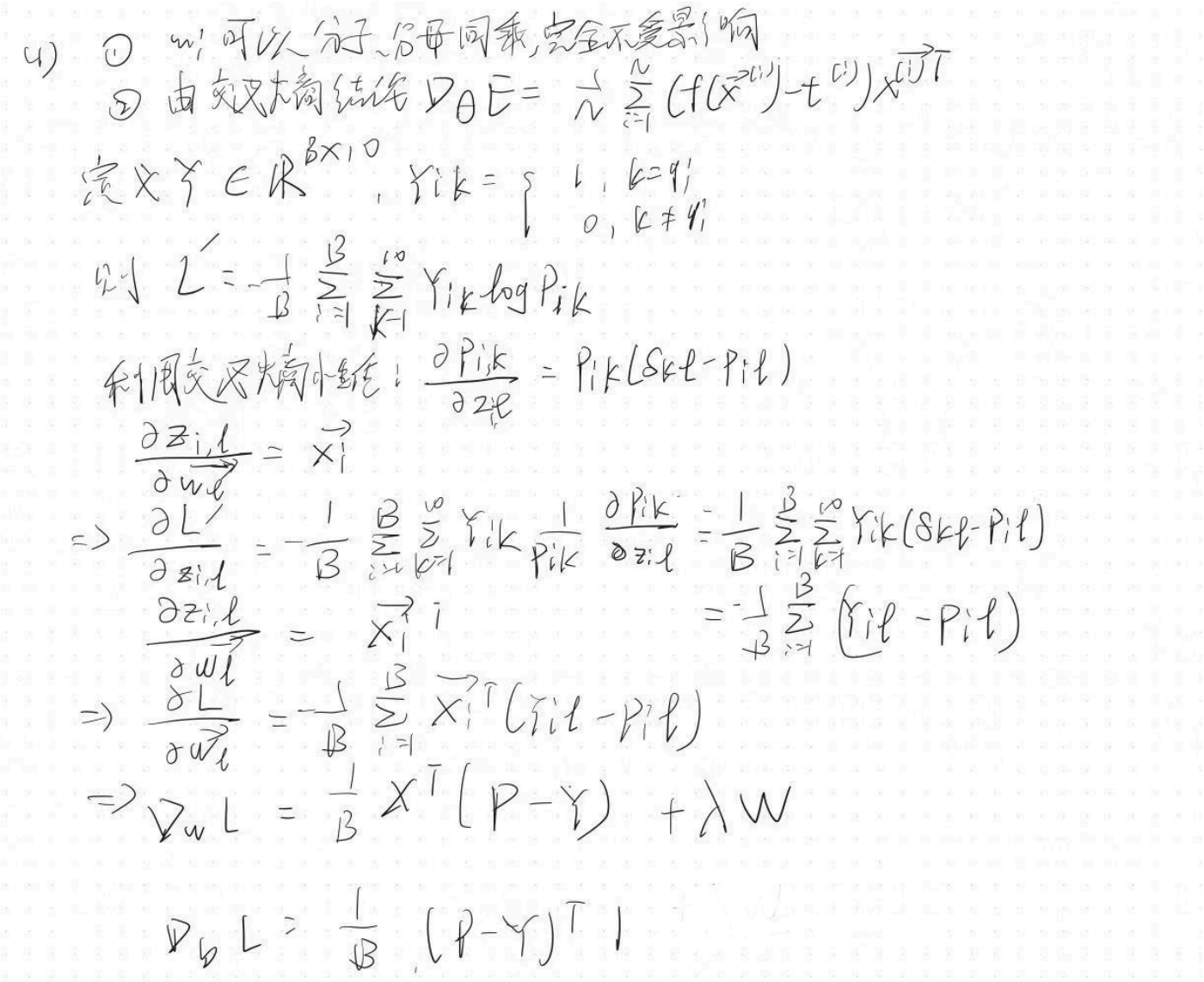
戴士博_2025012128_HW1_report

1.



加入m_i使得所有值被减去一个最大值，这使得所有指数都小于等于零。则所有e^x都小于一，从浮点数的表示来说，可以把**exp**表示的更加精准，且去除了溢出浮点数表示上限的风险，提升数值稳定性

2.Implement the classifier

```
Largest absolute gradient errors: 1.9761997593903402e-11 1.5758033766744006e-11
All checks passed.
```

3. Train and validate

```
Configuration: baseline
epoch  1/8: loss=0.4928, train acc=0.8761, val acc=0.8733
epoch  2/8: loss=0.4154, train acc=0.8907, val acc=0.8869
epoch  3/8: loss=0.3828, train acc=0.8969, val acc=0.8934
epoch  4/8: loss=0.3628, train acc=0.9015, val acc=0.8987
epoch  5/8: loss=0.3502, train acc=0.9036, val acc=0.8999
epoch  6/8: loss=0.3413, train acc=0.9068, val acc=0.9017
epoch  7/8: loss=0.3329, train acc=0.9088, val acc=0.9036
epoch  8/8: loss=0.3278, train acc=0.9106, val acc=0.9056

Configuration: smaller learning rate
epoch  1/8: loss=0.7460, train acc=0.8454, val acc=0.8418
epoch  2/8: loss=0.5764, train acc=0.8663, val acc=0.8628
epoch  3/8: loss=0.5078, train acc=0.8749, val acc=0.8718
epoch  4/8: loss=0.4687, train acc=0.8814, val acc=0.8779
epoch  5/8: loss=0.4432, train acc=0.8849, val acc=0.8812
epoch  6/8: loss=0.4246, train acc=0.8890, val acc=0.8863
epoch  7/8: loss=0.4103, train acc=0.8919, val acc=0.8881
epoch  8/8: loss=0.3992, train acc=0.8941, val acc=0.8912
```

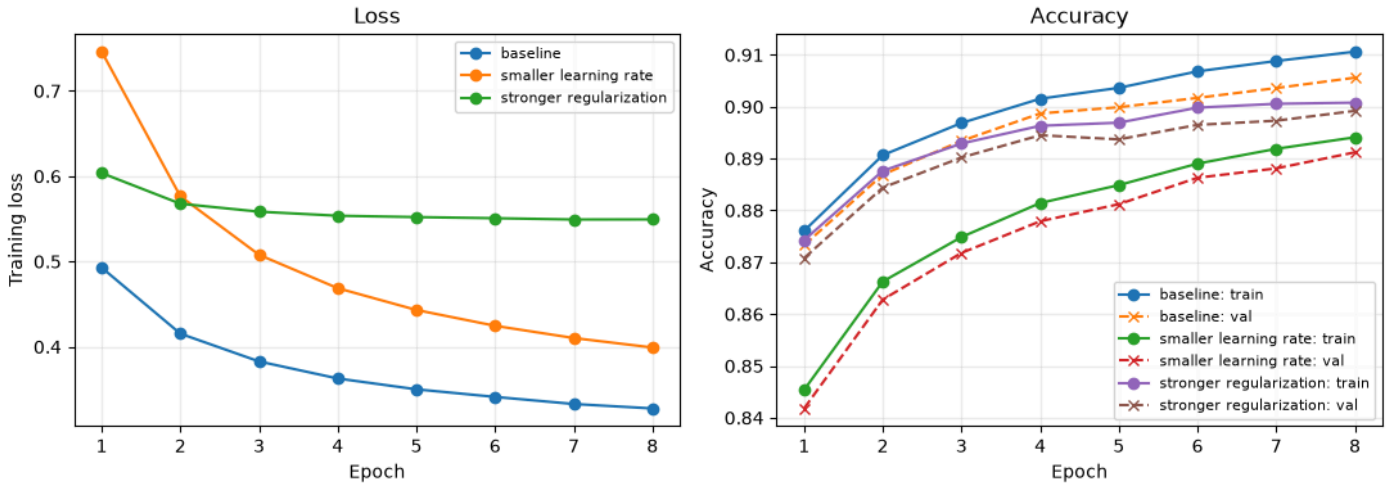
```
Configuration: stronger regularization
epoch  1/8: loss=0.6037, train acc=0.8743, val acc=0.8708
epoch  2/8: loss=0.5677, train acc=0.8876, val acc=0.8844
epoch  3/8: loss=0.5584, train acc=0.8929, val acc=0.8902
epoch  4/8: loss=0.5537, train acc=0.8963, val acc=0.8945
epoch  5/8: loss=0.5521, train acc=0.8969, val acc=0.8937
epoch  6/8: loss=0.5507, train acc=0.8998, val acc=0.8965
epoch  7/8: loss=0.5492, train acc=0.9006, val acc=0.8973
epoch  8/8: loss=0.5493, train acc=0.9008, val acc=0.8992

Results (copy to your report):
configuration                   lr     lambda    train acc    val acc
baseline                       0.1     0.0001       0.9106     0.9056
smaller learning rate         0.03     0.0001       0.8941     0.8912
stronger regularization        0.1       0.01       0.9008     0.8992
Selected by final validation accuracy: baseline
```

configuration	lr	λ	final train acc	final val acc
baseline	0.10	1×10−4	0.9106	0.9056
smaller learning rate	0.03	1×10−4	0.8941	0.8912

configuration	lr	λ	final train acc	final val acc
stronger regularization	0.10	1×10−2	0.9008	0.8992

最终根据valid accuracy选择了baseline



4.Evaluate and discuss

baseline的test表现

```
Selected configuration: baseline
Final validation accuracy: 0.9056
Test accuracy: 0.9143
```

学习率主要影响收敛速度：baseline（η=0.10）的训练损失从 0.4928 快速降到 0.3278，而 η=0.03 那组到第 8 轮仍有 0.3992、曲线斜率未见明显放缓，最终训练/验证准确率分别低了约 1.7 和 1.4 个百分点（0.9106/0.9056 对 0.8941/0.8912），说明在固定 8 个 epoch 下学习率偏小会造成欠训练。正则化的影响方向相反但幅度更小：把 λ从 1×10−41×10−4 增大到 1×10−21×10−2 后，损失在第 4 轮就停在约 0.55 不再下降，训练与验证准确率都略低于 baseline（0.9008/0.8992），说明过强的 L2 抑制了权重增长，把模型压得更欠拟合。三组配置的训练与验证准确率差距都不超过 0.6 个百分点，验证准确率也没有随训练回落，因此没有过拟合迹象；相反 baseline 的训练损失到第 8 轮仍在下降，整体属于欠拟合。据此按验证准确率选定 baseline（0.9056），一次性测试准确率为 0.9143，略高于验证结果属于正常波动。

线性像素分类器的一项根本局限是对每个类别只学一张 28×28 的模板（权重图），无法建模像素之间的空间关系，因此对平移、旋转和笔画粗细变化都很敏感，这也是 4/9、3/5、7/1 这类形状相近数字容易混淆的原因。